

Peiyang Song

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📁 [peiyang-song.github.io](https://github.com/peiyang-song)

Education

- 5/2031 **Carnegie Mellon University** Pittsburgh, PA
(expected) *PhD in Language and Information Technology*
- 6/2026 **California Institute of Technology** Pasadena, CA
(expected) *BS in Computer Science & Minor in Robotics*
Advisors: Prof. Steven Low & Prof. Günter Niemeyer. GPA: **4.2/4.0**.

Work Experience

- 6/2026–8/2026 **Apodex** Redwood City, CA
Research Intern, Verifiable AI Lab. Mentor: Dr. Kaiyu Yang.

Research Interests

Formal Reasoning · Natural Language Reasoning · Neuro-Symbolic AI · AI4MATH · LLMs

Selected Publications & Preprints

Full list on Google Scholar: <https://scholar.google.com/citations?user=E1j11NQAAAAJ>

- Preprint **Interactive Evaluation Requires a Design Science**
Keyang Xuan*, Peiyang Song*, Pan Lu, Pengrui Han, Wenkai Li, Zhenyu Zhang, Zexue He, Wenyue Hua, Manling Li, Jiaxuan You, Adrian Weller, Yizhong Wang[†], and Jiaxin Pei[†] (* Equal Contribution, [†] Equal Advising)
ICML Workshop on Combining Theory and Benchmarks (CTB), 2026
- Preprint **Characterizing Initial Human-AI Proof Formalization Workflows**
Katherine M. Collins*, Simon Frieder*, Jonas Bayer, Jacob Loader, Jeck Lim, Peiyang Song, Fabian Zasier, Lexin Zhou, Shanda Li, Sam Looi, Jose Hernandez-Orallo, Joshua B. Tenenbaum, Cameron Freer, Umang Bhatt, Adrian Weller, Valerie Chen[†], Ilia Sucholutsky[†] (* Equal Contribution, [†] Equal Advising)
NeurIPS Workshop on Mathematical Reasoning and AI (MATH-AI), 2025
- Preprint **Adaptation of Agentic AI: A Survey of Post-Training, Memory, and Skills**
Pengcheng Jiang*, Jiacheng Lin*, Zhiyi Shi*, Zifeng Wang, Luxi He, Yichen Wu, Ming Zhong, Peiyang Song, Qizheng Zhang, Heng Wang, Xueqiang Xu, Hanwen Xu, Pengrui Han, Dylan Zhang, Jiashuo Sun, Chaoqi Yang, Kun Qian, Tian Wang, Changran Hu, Manling Li, Quanzheng Li, Hao Peng, Sheng Wang, Jingbo Shang, Chao Zhang, Jiaxuan You, Liyuan Liu, Pan Lu, Yu Zhang, Heng Ji, Yejin Choi, Dawn Song, Jimeng Sun, Jiawei Han (* Equal Contribution)

Preprint, 2025

Preprint **How and Why LLMs Generalize: A Fine-Grained Analysis of LLM Reasoning from Cognitive Behaviors to Low-Level Patterns**

Haoyue Bai, Yiyao Sun, Wenjie Hu, Shi Qiu, Maggie Ziyu Huan, Peiyang Song, Robert Nowak, Dawn Song

Preprint, 2025

TMLR 2026 **Large Language Model Reasoning Failures**

Peiyang Song^{*}, Pengrui Han^{*}, Noah Goodman (* Equal Contribution)

Transactions on Machine Learning Research, 2026, Survey Certification

ICML 2026 **The Personality Illusion: Revealing Dissociation Between Self-Reports & Behavior in LLMs**

Pengrui Han^{*}, Rafal D. Kocielnik^{*}, Peiyang Song, Ramit Debnath, Dean Mobbs, Anima Anandkumar, R. Michael Alvarez (* Equal Contribution)

International Conference on Machine Learning, 2026

NeuS 2025 **Lean Copilot: Large Language Models as Copilots for Theorem Proving in Lean**

Peiyang Song, Kaiyu Yang, Anima Anandkumar

International Conference on Neuro-symbolic Systems, 2025, 1.3k+ stars on Github, ranking 2nd after Mathlib4 among all Lean projects

IEEE Micro 2025 **Delay Space Arithmetic and Architecture**

Rhys Gretsch, Peiyang Song, Advait Madhavan, Jeremy Lau, Timothy Sherwood

IEEE Micro, 2025, Top Pick Award

ICLR 2025 **LeanAgent: Lifelong Learning for Formal Theorem Proving**

Adarsh Kumarappan^{*}, Mo Tiwari^{*}, Peiyang Song, Robert Joseph George, Chaowei Xiao, Anima Anandkumar

International Conference on Learning Representations, 2025

TMLR 2025 **LeanProgress: Guiding Search for Neural Theorem Proving via Proof Progress Prediction**

Robert Joseph George, Suozhi Huang, Peiyang Song, Anima Anandkumar

Transactions on Machine Learning Research, 2025

EMNLP 2024 **In-Context Learning May Not Elicit Trustworthy Reasoning: A-Not-B Errors in Pretrained Language Models**

Pengrui Han^{*}, Peiyang Song^{*}, Haofei Yu, Jiaxuan You (* Equal Contribution)

Conference on Empirical Methods in Natural Language Processing: Findings, 2024

EMNLP 2024 **Creative and Context-Aware Translation of East Asian Idioms with GPT-4**

Kenan Tang^{*}, Peiyang Song^{*}, Yao Qin, Xifeng Yan (* Equal Contribution)

Conference on Empirical Methods in Natural Language Processing: Findings, 2024

ASPLOS 2024 **Energy Efficient Convolution with Temporal Arithmetic**

Rhys Gretsch, Peiyang Song, Advait Madhavan, Jeremy Lau, Timothy Sherwood
ACM International Conference on Architectural Support for Programming Languages and Operating Systems, 2024

NeurIPS 2023 **LeanDojo: Theorem Proving with Retrieval-Augmented Language Models**
Kaiyu Yang, Aidan Swope, Alex Gu, Rahul Chalamala, Peiyang Song, Shixing Yu, Saad Godil, Ryan Prenger, Anima Anandkumar
Conference on Neural Information Processing Systems, 2023, Oral Presentation

Selected Academic Awards

- 5/2026 **Outstanding Reviewer Award @ EACL**
In recognition of outstanding review service at EACL 2026
- 1/2026 **Survey Certification @ TMLR**
For paper “Large Language Model Reasoning Failures”
- 12/2025 **Best Paper Honorable Mention Award @ NeurIPS LAW Workshop**
For paper “The Personality Illusion: Revealing Dissociation Between Self-Reports & Behavior in LLMs”
- 5/2025 **ICLR Notable Reviewer Award**
In recognition of outstanding review service at ICLR 2025
- 4/2025 **George W. Housner Student Discovery Fund**
For paper “Lean Copilot: Large Language Models as Copilots for Theorem Proving in Lean”
- 2/2025 **IEEE Micro Top Pick Award**
For paper “Energy Efficient Convolutions with Temporal Arithmetic”
- 4/2023 **Caltech SURF Fellowship**
For research work on machine learning for theorem proving in Lean

Selected Media

- 2026 **'Not how you build a digital mind': How reasoning failures are preventing AI models from achieving human-level intelligence**
Live Science
- 2026 **Scientists Found AI's Fatal Flaw—The Most Advanced Models Are Failing Basic Logic Tests**
Popular Mechanics
- 2025 **New Framework Simplifies the Complex Landscape of Agentic AI**
VentureBeat
- 2025 **This AI Paper Explains Why Most “Agentic AI” Systems Feel Impressive in Demos and then Completely Fall Apart in Real Use**
MarkTechPost

- 2025 **Researchers Discover “Personality Illusion” to Reveal a Profound Disconnect Between Language and Behavior in LLMs**
MIT Technology Review China
- 2024 **Mathematicians’ Newest Assistants Are Artificially Intelligent**
Scientific American
- 2024 **LeanAgent: The First Life-Long Learning Agent for Formal Theorem Proving in Lean**
MarkTechPost
- 2024 **Lean Copilot: An AI Tool That Allows Large Language Models (LLMs) to Be Used in Lean for Proof Automation**
MarkTechPost
- 2023 **Can LLMs Generate Mathematical Proofs That Can Be Rigorously Checked?**
MarkTechPost

Invited Talks & Tutorials

Large Language Model Reasoning Failures

- 3/2026 Fireside Chat @ Ploutos.ai
- 3/2026 BlueDot AI Safety Reading Group

Tutorial: Neuro-Symbolic Theorem Proving with Lean

- 9/2024 3rd Neuro-Symbolic AI Summer School (NSSS)

Towards An AI Mathematician

- 12/2023 UC Santa Barbara NLP Lab
- 11/2023 CCS Research & Creative Activities Conference (RACA-CON)
- 8/2023 Caltech SURF Seminar Day

Teaching Experience

- Spring 2026 **CS 38/138: Algorithms**
Teaching Assistant @ *California Institute of Technology*
- Winter 2026 **ME/CS/EE 133B: Robotics – Planning**
Teaching Assistant @ *California Institute of Technology*
- Fall 2025 **ME/CS/EE 133A: Robotics – Kinematics**
Teaching Assistant @ *California Institute of Technology*

Academic Services

- Reviewer** Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- Conference on Language Modeling (COLM)
- Association for Computational Linguistics Rolling Review (ARR)

Annual Meeting of the Association for Computational Linguistics (ACL)
Conference on Empirical Methods in Natural Language Processing (EMNLP)
International Joint Conference on Natural Language Processing (IJCNLP)
Asia-Pacific Chapter of the Association for Computational Linguistics (AACL)

NeurIPS Mathematical Reasoning and AI (MATH-AI) Workshop
NeurIPS Workshop on Deep Learning for Code (DL4C)
NeurIPS Workshop on Behavioral Machine Learning
ICLR VerifAI: AI Verification in the Wild Workshop
ICLR Workshop on Representational Alignment (Re-Align)
ICML AI for Math (AI4MATH) Workshop
ICML Workshop on LLMs and Cognition (LLM-Cognition)
ICML Workshop on Assessing World Models
ICML Workshop on Models of Human Feedback for AI Alignment (MoFA)

Organizing Staff Agentic AI Summit 2025 @ UC Berkeley

On-Campus Services & Appointments

Caltech Admissions Ambassador @ Caltech Undergraduate Admissions Office
First-Year Caltech Connector (FCC) @ Student & Family Engagement Office